

- Q.29 Define resolved motion control and its role in manipulator control
- Q.30 Summarize the main differences between manual teaching and lead through programming.
- Q.31 State two features of VAN that make it suitable for industrial robots.
- Q.32 Describe key functions performed by robot programming.
- Q.33 Explain benefits of using robots rather than humans for industrial inspection.
- Q.34 What does a multi-axis painting robot enable that simple robots cannot?
- Q.35 Describe key advantages of using robots for material transfer operations.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Define a robot and explain its main components with working of each.
- Q.37 Explain the term robot programming language and its types. How is it different from normal computer languages.
- Q.38 Explain the concept of new minimum time control in robotics. How does this control strategy optimize robot operations?

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**6th Sem /
Mechatronics / Mech. (CAD & CAM Design & Robotics)
Sub : Robotics**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Robots are used in :
a) Dangerous environments
b) Only outdoors
c) Only in homes
d) Only for sports
- Q.2 Closed loop robots rely on :
a) Feedback from sensors
b) Human control only
c) Open loop commands
d) None of the above
- Q.3 The working principle of pneumatic gripper is based on :
a) Electric motors b) Hydraulic power
c) Compressed air d) Magnetism
- Q.4 Which drive type is best for fast and accurate movements with low load?
a) Pneumatic b) Hydraulic
c) Electric d) Manual
- Q.5 Which control method specifically improves performance under changing conditions?
a) Adaptive control
b) open loop control

- c) Mechanical control only
 - d) Manual control
- Q.6 New minimum time control is focussed on :
- a) Maximizing delay
 - b) Minimizing the time required for robot movements
 - c) Ignoring time in control
 - d) Manual override
- Q.7 The simplest method of robot programming is :
- a) Manual teaching
 - b) Lead through programming
 - c) C/C++ programming
 - d) Simulation
- Q.8 Robot programming allows robots to :
- a) Sleep during operation
 - b) Interpret sensor data and actuate end-effectors
 - c) Avoid all human interaction
 - d) Fly
- Q.9 Pick-and place robots are an example of :
- a) Material transfer applications
 - b) Painting robots
 - c) Welding robots
 - d) None of these
- Q.10 Key advantage of robotic painting is :
- a) Uneven coating
 - b) Uniform and consistent coverage
 - c) Increased paint waste
 - d) Higher human exposure to chemicals

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The tool fitted at the end of a robot's arm is called the _____.

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- Q.12 The full form of LVDT is _____.
- Q.13 Servo motors provide precise _____ and speed control.
- Q.14 The number of independent motions a robot can make is called its _____.
- Q.15 Servo control systems generally provide better accuracy than non-servo systems. (True/False)
- Q.16 Python in robotics is unsuitable for AI integration. (True/False)
- Q.17 State two main functions of robot programming.
- Q.18 What is meant by offline programming in robotics?
- Q.19 What are inspection robots used for?
- Q.20 Mention one advantage of robotic painting.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe a situation where a robot can perform a task that is too dangerous for humans
- Q.22 Why is regular maintenance required for robots?
- Q.23 What are the advantages of using articulated robots over other types?
- Q.24 What are the typical applications of hydraulic drives in robotics?
- Q.25 What are RTD sensor, and why are they important in robotics?
- Q.26 What are the key criteria to consider when selecting a robot for an industrial application?
- Q.27 Why is it important to have multiple control levels in robotic systems?
- Q.28 What is adaptive control in robotics, and why is it used?

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